

host_agent

User: Find all buildings in Exeter

planning_agent

- 1. Given the query, I will read and understand it.
- 2. I will make a number of steps required to solve the query.
- 3. For this query, the thought process is as follows:
 - a. Identify the general geographical area of the query, which is Exeter (1 entry).
 - b. The entities to be searched within this geographical area are buildings.
 - c. There are no specific conditions mentioned in the query, so I will aim to find all buildings in Exeter.
 - d. Information about the count of entities is not required in this case.

output steps: ["Find Exeter 1 area", "Search for all buildings in Exeter as many search results as possible"]

named_area

User: Find the area polygon for Exeter.

Tool call_os_ngd with args: - filters: None - bbox: None - polygon_or_point: True - street_address: None - filename: 'exeter_polygon'

Tool call_os_ngd returned result: - Search results have been found. - Artifact generated is: exeter_polygon - Description: A geopandas dataframe containing named area data with no filters applied for the query (so remember no search is performed here so spurious entities present) Find the area polygon for Exeter. - Count of records fetched: 98

data_analysis_agent

User: Filter the artifact 'exeter_polygon' to find the specific polygon for Exeter. Save the filtered results as 'filtered_exeter_polygon'.

Tool generate_metadata_for_artifacts with args: - artifact_names: ['exeter_polygon']

Tool generate_metadata_for_artifacts returned result: (metadata for 'exeter_polygon')

Tool Code Executor

```
def filter_exeter_polygon(data: list):
    gdf = data[0]
    # Filter using multiple columns to ensure we get the polygon for Exeter
    # We'll check 'name1_text', 'description', and 'theme' for relevant values
    filtered = gdf[
        (gdf['name1_text'].str.contains('Exeter', case=False, na=False)) |
        (gdf['description'].str.contains('Exeter', case=False, na=False)) |
        (gdf['theme'].str.contains('Exeter', case=False, na=False))
    ]
    return [
        f"Filtered to {len(filtered)} polygons for Exeter.",
        'filtered_exeter_polygon',
        'A GeoDataFrame containing only the polygons specifically related to Exeter, filtered using name, description, and theme columns.',
        filtered
    ]
```

Tool code_executor returned result: ['Filtered to 1 polygons for Exeter.',]

Output: Filtered to 1 polygon for Exeter. An artifact has been generated with the name 'filtered_exeter_polygon' and contains a GeoDataFrame specifically related to the Exeter polygon, filtered using the name, description, and theme columns.

named_area

Output: The specific polygon for Exeter has been filtered and saved as 'filtered_exeter_polygon'. This artifact contains only the polygons specifically related to Exeter.

buildings_agent

User: Search for all buildings within the area defined by the artifact 'filtered_exeter_polygon'.

Tool call_os_ngd with args: - filters: None - bbox: 'filtered_exeter_polygon' - polygon_or_point: True - street_address: False - filename: 'exeter_buildings'

Tool call_os_ngd returned result: - Multiple search results have been found in multiple datasets. A summary of each is provided. - Artifacts generated are: ['buildings_exeter_buildings', 'buildingline_exeter_buildings', 'buildingpart_exeter_buildings'] - Descriptions: ["A geopandas dataframe containing building data with filters and bbox applied for the query Search for all buildings within the area defined by the artifact 'filtered_exeter_polygon'. using filters None.",

"A geopandas dataframe containing building line data with filters and bbox applied for the query Search for all buildings within the area defined by the artifact 'filtered_exeter_polygon'. using filters None.", "A geopandas dataframe containing building part data with filters and bbox applied for the query Search for all buildings within the area defined by the artifact 'filtered_exeter_polygon'. using filters None."] - Counts of records fetched are: [69169, 69342, 85480]

Output: The search for all buildings within the area defined by the artifact 'filtered_exeter_polygon' has returned the following artifacts:

1. **buildings_exeter_buildings**: A geopandas dataframe containing building data. (69169 records)
2. **buildingline_exeter_buildings**: A geopandas dataframe containing building line data. (69342 records)
3. **buildingpart_exeter_buildings**: A geopandas dataframe containing building part data. (85480 records)

These artifacts contain comprehensive building data for the specified area. If you need further analysis or filtering, please let me know!

plotting_agent

User: Plot all buildings in Exeter using the artifacts 'buildings_exeter_buildings', 'buildingline_exeter_buildings', and 'buildingpart_exeter_buildings' within the area defined by 'filtered_exeter_polygon'. Provide a summary of the results.

Tool generate_metadata_for_artifacts with args: - artifact_names: ['buildings_exeter_buildings', 'buildingline_exeter_buildings', 'buildingpart_exeter_buildings', 'filtered_exeter_polygon']

Tool generate_metadata_for_artifacts returned result: (metadata for the artifacts)

Tool Code Executor

```
def plot_exeter_buildings(data: list):
    import geopandas as gpd
    import folium
    from shapely.geometry import mapping
    import pandas as pd

    # Unpack data
    buildings = data[0].copy()
    building_lines = data[1].copy()
    building_parts = data[2].copy()
    exeter_poly = data[3].copy()

    # Ensure all are in EPSG:4326
    buildings = buildings.to_crs(epsg=4326)
```

```

building_lines = building_lines.to_crs(epsg=4326)
building_parts = building_parts.to_crs(epsg=4326)
exeter_poly = exeter_poly.to_crs(epsg=4326)

# Remove problematic columns (e.g., Timestamp) for folium plotting
def remove_timestamps(gdf):
    for col in gdf.columns:
        if pd.api.types.is_datetime64_any_dtype(gdf[col]):
            gdf[col] = gdf[col].astype(str)
    return gdf
buildings = remove_timestamps(buildings)
building_lines = remove_timestamps(building_lines)
building_parts = remove_timestamps(building_parts)
exeter_poly = remove_timestamps(exeter_poly)

# Get the Exeter polygon (assuming single row)
exeter_geom = exeter_poly.iloc[0].geometry

# Filter all buildings, lines, and parts within Exeter polygon
buildings_in_exeter = buildings[buildings.intersects(exeter_geom)]
building_lines_in_exeter = building_lines[building_lines.intersects(exeter_geom)]
building_parts_in_exeter = building_parts[building_parts.intersects(exeter_geom)]

# Create folium map centered on Exeter polygon centroid
centroid = exeter_geom.centroid
m = folium.Map(location=[centroid.y, centroid.x], zoom_start=15)

# Add Exeter polygon
folium.GeoJson(mapping(exeter_geom), name='Exeter Area', style_function=lambda x: {'color': 'blue', 'weight': 2, 'fillOpacity': 0.05}).add_to(m)

# Plot buildings
folium.GeoJson(buildings_in_exeter, name='Buildings', style_function=lambda x: {'color': 'red', 'weight': 1, 'fillOpacity': 0.3}).add_to(m)
# Plot building lines
folium.GeoJson(building_lines_in_exeter, name='Building Lines', style_function=lambda x: {'color': 'green', 'weight': 2}).add_to(m)
# Plot building parts
folium.GeoJson(building_parts_in_exeter, name='Building Parts', style_function=lambda x: {'color': 'orange', 'weight': 1, 'fillOpacity': 0.2}).add_to(m)

folium.LayerControl().add_to(m)

# Save map
map_filename = 'exeter_buildings_map.html'
m.save(map_filename)

# Prepare summary
summary = f"Buildings in Exeter: {len(buildings_in_exeter)} | Building lines: {len(building_lines_in_exeter)} | Building parts: {len(building_parts_in_exeter)}"
summary += '\nSample buildings:\n' + str(buildings_in_exeter[['osid', 'description', 'geometry_area_m2']].head(5))
summary += '\nSample building lines:\n' + str(building_lines_in_exeter[['osid', 'description', 'geometry_length']].head(5))
summary += '\nSample building parts:\n' + str(building_parts_in_exeter[['osid', 'description', 'geometry_area_m2']].head(5))

```

```

    artifact_description = (
        "Folium map showing all buildings, building lines, and building parts within the Exeter area polygon. "
        "Buildings are shown in red, building lines in green, and building parts in orange. The Exeter area boundary is outlined in blue. "
        "The map allows visual inspection of the spatial distribution of these features within Exeter."
    )

    return [summary, 'exeter_buildings_map', artifact_description, map_filename]
```

Tool code_executor returned result:

- Buildings in Exeter: 69169 | Building lines: 69342 | Building parts: 85480

Sample buildings:

	osid	description \
45098	32ff414d-d1c8-454b-9b67-ac472217e4ce	Domestic Outbuilding
62699	b9a49c80-fecf-4308-a98d-cf90ea4ea2e4	End-Of-Terrace House
24664	9dfb4b41-f53d-44c0-8a7a-5f7884375677	Mid-Terrace House
27514	aea058f2-6a39-4829-ae6b-85c6863a250e	Mid-Terrace House
37791	ef3285ad-8a1d-4d4d-aa50-333888085e10	Unknown Building

	geometry_area_m2
45098	18.438
62699	99.338
24664	78.565
27514	66.910
37791	7.858

Sample building lines:

	osid	description \
24048	57202087-5453-473f-b927-0b58b9e564fb	Building Internal Division
59314	d87c4727-0467-40c3-b148-1fae6bef930c	Building Occupier Division
15541	38e4a6bd-c82d-4fe5-908f-df1497b7a199	Building Occupier Division
20661	4ad601b7-2d43-4ffd-a813-785bdfac9349	Building Occupier Division
22011	4fc5cd11-808d-4791-8676-b52213bce65b	Building Occupier Division

	geometry_length
24048	5.254560
59314	8.897830
15541	10.000591
20661	9.427393
22011	10.000591

Sample building parts:

	osid	description	geometry_area_m2
73630	c7f949ce-e1f7-425d-b6fa-8ddc11065214	Building	20.297
46871	7436b5c6-ff93-41cb-aeb1-c3c22acaf011	Building	16.914
46518	6fcc76f4-817e-46e4-92c7-6bd50bd9c52d	Building	8.207
56955	90474fb7-6565-4104-ba7b-21f7fe084140	Building	54.607
74605	cd808371-63d1-49b0-993e-87397a2aef3d	Building	8.505

Artifact generated: 'exeter_buildings_map' (Folium map showing all buildings, building lines, and building parts within the Exeter area polygon. Buildings are shown in red, building lines in green, and building parts in orange. The Exeter area boundary is outlined in blue. The map allows visual inspection of the spatial distribution of these features within Exeter.)

host_agent

All buildings in Exeter have been found and mapped. Here is a summary of the results:

- Buildings: 69,169
- Building lines: 69,342
- Building parts: 85,480

Sample building types include domestic outbuildings, end-of-terrace houses, mid-terrace houses, and unknown buildings. The data includes both the polygons of the buildings and their internal divisions and parts.

A map has been generated showing: - All buildings in red - Building lines in green - Building parts in orange - The Exeter area boundary in blue

This map allows you to visually inspect the spatial distribution of all buildings within Exeter. If you need the map file or further analysis, let me know!